

BAD714B Business Analytics TIE SIMP questions by Prof Manjunath

Module 1

1. Define the concept of Business Intelligence and analyze how it facilitates effective decision-making in a corporate setting
2. Trace the historical evolution of computerized systems designed for decision support from their inception to the present day
3. Provide a detailed comparison between Business Intelligence, Data Science, and Artificial Intelligence, highlighting their distinct roles
4. Discuss the specific shifts in the global business environment that are currently forcing organizations to adopt advanced analytics
5. Distinguish between Traditional AI and Conversational AI, explaining the technological drivers behind each
6. Using relevant industry examples, evaluate how Artificial Intelligence is currently transforming traditional business operations
7. Identify the primary structural and technical challenges organizations encounter when implementing a BI framework

Module 2

1. Explain the taxonomy of data and provide practical examples for each category found in business applications
2. Discuss the importance of data preprocessing, explaining why it is often described as both an art and a science
3. Define Big Data and elaborate on its fundamental characteristics, specifically the 5Vs¹⁰.
4. Explain the concept of Stream Analytics and discuss its significance in handling real-time data environments
5. Differentiate between descriptive and inferential statistics, explaining their respective applications in business analytics
6. What is regression modeling? Provide a clear example of how it can be used to solve a business problem

7. Examine the specific complexities and challenges involved in the preprocessing phase of Big Data analytics

Module 3

1. Describe the standard architecture of a Business Intelligence system and explain the function of its various components
2. Detail the essential phases involved in the systematic development of a Data Warehouse
3. Analyze different Data Warehouse architectures and illustrate your answer with appropriate structural diagrams
4. Compare and contrast OLAP and OLTP systems, focusing on their operational requirements and data structures
5. Evaluate the various types of charts and graphs used in data visualization, specifying which are most effective for different data scenarios.
6. Define Information Dashboards and explain the key features that make them vital for executive-level decision support.
7. Discuss the limitations and risks associated with relying on traditional, legacy BI approaches in the modern digital era.

Module 4

1. Define Data Mining and outline its primary objectives within a commercial context.
2. Explain the step-by-step process of a standard data mining project, from data collection to deployment.
3. Distinguish between supervised and unsupervised learning, providing examples of algorithms used for each.
4. Describe the mathematical structure of decision support models and how they represent real-world business problems.
5. Discuss the role of optimization in prescriptive analytics and how it helps in maximizing organizational outcomes.

6. Explain how decision-makers categorize environments based on certainty, uncertainty, and risk in analytical modeling.
7. What is the role of simulation in decision-making? Explain how it provides a safe environment for testing business strategies.

Module 5

1. Define Text Analytics and discuss its diverse applications in understanding unstructured business data.
2. Examine the role of Natural Language Processing (NLP) as a foundational technology for effective Text Mining
3. Explain the detailed process of Text Mining and how it differs from traditional data mining.
4. What is Sentiment Analysis? Provide examples of how businesses use it to gauge customer opinions from social media.
5. Explain the concept of Web Mining and differentiate between Web Content Mining and Web Usage Mining.
6. Define Social Network Analysis and explain how centrality measures help in identifying key influencers within a network.
7. Discuss the ethical considerations and privacy issues that arise when performing text and web mining on public data.